

Ethical Hacking

Complete student lesson for course code **5133B**.

Revision 2026 Semester 5 Program Elective 4 modules

Course snapshot

Scheme: 4 (L:4 T:0 P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Handbook method: Do not treat a topic as completed after reading its heading. For every topic card, complete the learning objectives, follow the conceptual framework, write the worked answer method, and answer the self-check questions. This creates a study record that is useful for class learning and examination preparation.

Course Dashboard

Course code

5133B

Semester

5

Credits

4

Teaching scheme

4 (L:4 T:0 P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Hacking, Malware, Network Attacks and Countermeasures	14	CO1
2	Footprinting, Social Engineering and Port Scanning	16	CO2

No.	Module	Official hours	Relevant CO / level
3	Operating-System Vulnerabilities and Hardening	15	CO3
4	Web-Application and Wireless-Hacking Countermeasures	13	CO4

Prerequisites

- Digital fundamentals — Digital Computer Fundamentals, semester 3.
- Basic programming concepts — Programming in C, semester 3.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. Impart the concepts in ethical hacking.
2. Introduce various network and system attacks, countermeasures and tools involved.
3. Provide knowledge about current security issues in computer networks.

Course Outcomes

1. CO1: Summarize the concepts of hacking, Malwares, Network attacks and counter measures — 14 hours — Understanding.
2. CO2: Demonstrate foot printing and port scanning techniques using simple tools — 16 hours — Understanding.
3. CO3: Summarize operating systems and web Application Vulnerabilities — 15 hours — Understanding.

4. CO4: Outline counter measures for web application and wireless hacking — 13 hours — Understanding.

CO-PO mapping as supplied

CO1: PO1 = 2 and PO6 = 2 (Moderately mapped).

CO2: PO1 = 2 and PO6 = 2 (Moderately mapped).

CO3: PO1 = 2 and PO6 = 2 (Moderately mapped).

CO4: PO1 = 2 and PO6 = 2 (Moderately mapped).

Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

Source-to-lesson coverage map

Module	Lesson topic cards	Source basis	Official point units
Module 1	5	Official topic-row context	5
Module 2	4	Official topic-row context	4
Module 3	4	Official topic-row context	4
Module 4	4	Official topic-row context	4

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Hacking and ethical hacking	2
module-1	M1.02	Ethical issues related to hacking	2

Module	Topic code	Topic	Hours
module-1	M1.03	Malware classes, actions and protection	4
module-1	M1.04	Network attacks	4
module-1	M1.05	Countermeasures for network attacks	2
module-2	M2.01	Footprinting and tools used for it	3
module-2	M2.02	Competitive intelligence methods	3
module-2	M2.03	Social-engineering methods	4
module-2	M2.04	Port scanning, methods and tools	6
module-3	M3.01	Windows vulnerable components	6
module-3	M3.02	Methods for hardening Windows	3
module-3	M3.03	Linux OS vulnerabilities	3
module-3	M3.04	Countermeasures against Linux attacks	3
module-4	M4.01	Web applications and components	3
module-4	M4.02	Web-application vulnerabilities and countermeasures	5
module-4	M4.03	Wireless network components and their use	2
module-4	M4.04	Countermeasures for wireless attacks	3

Learning Roadmap

1. Hacking, Malware, Network Attacks and Countermeasures
CO1 · 14 hours

2. Footprinting, Social Engineering and Port Scanning
CO2 · 16 hours

3. Operating-System Vulnerabilities and Hardening
CO3 · 15 hours

4. Web-Application and Wireless-Hacking Countermeasures
CO4 · 13 hours

The roadmap follows the order of the supplied syllabus.

Hacking, Malware, Network Attacks and Countermeasures

Ethical hacking studies security weaknesses only with permission, defined scope, controlled methods, and responsible reporting. This module establishes the vocabulary of threats, malware, attacks, and the CIA triad.

Hours: 14

CO1 · Bloom level: Understanding

Handbook study routine: Complete Module 1 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 1

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding

M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding.

M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network

M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks

M1.05 2 Understanding attacks Contents:

Point-by-point study guide

– **Official syllabus point 1: M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding**

Exact source point

Syllabus text: M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding

How to understand and study it

This point is an official syllabus requirement: “M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ് ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding using the official terminology retained above.
- Organise the important elements of M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding?
- How would you distinguish M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding from a closely related idea?
- Where would an engineer or technician encounter M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.01 2 Understanding. Hacking. Identify various ethical issues related to M1.02 2 Understanding താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term നൽകുക. താഴെ പറയുന്ന definition നൽകുക principle, named parts/steps, application, limitation നൽകുക. English terminology, symbols, units, diagram labels നൽകുക. Exam answer concept-കൾക്ക് ഉദാഹരണങ്ങൾ നൽകുക.

– **Official syllabus point 2: M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding.**

Exact source point

Syllabus text: M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding.

How to understand and study it

This point is an official syllabus requirement: “M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding.”. Prepare a classification table. For every named type, learn its defining feature, distinguishing point, and one appropriate engineering context. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Prepare a table with type, defining feature, advantage or limitation, and application.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.02 2 Understanding hacking Classify different malwares, their actions, M1.03 4 Understanding. ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. using the official terminology retained above.
- Organise the important elements of M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. into a logical sequence, classification, structure, or calculation.
- Connect M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding.?
- How would you distinguish M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. from a closely related idea?
- Where would an engineer or technician encounter M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding., and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.

- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.02 2 Understanding hacking Classify different malwares, their actions and M1.03 4 Understanding. **English** topic **Malayalam** exact technical term **English** definition **Malayalam** principle, named parts/steps, application, limitation **English** **Malayalam** **English** terminology, symbols, units, diagram labels **Malayalam** **English**. Exam answer **Malayalam** concept-**English** **Malayalam**-**English** **Malayalam** **English**.

– **Official syllabus point 3: M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network**

Exact source point

Syllabus text: M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network

How to understand and study it

This point is an official syllabus requirement: “M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network”. Record the relevant quantity, symbol, unit, relation or formula, and the interpretation of the result. Do not memorise a number without its unit and condition. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Record the symbol, SI unit, governing relation, and one correctly labelled calculation.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network using the official terminology retained above.
- Organise the important elements of M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network into a logical sequence, classification, structure, or calculation.
- Connect M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data

item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network?
- How would you distinguish M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network from a closely related idea?
- Where would an engineer or technician encounter M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M1.03 4 Understanding. methods for protection. M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network
 topic exact technical term
 definition principle, named parts/steps, application, limitation
 English terminology, symbols, units, diagram labels
 Exam answer
 concept-

– Official syllabus point 4: M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks

Exact source point

Syllabus text: M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks

How to understand and study it

This point is an official syllabus requirement: “M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks”. Record the relevant quantity, symbol, unit, relation or formula, and the interpretation of the result. Do not memorise a number without its unit and condition. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Record the symbol, SI unit, governing relation, and one correctly labelled calculation.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks using the official terminology retained above.
- Organise the important elements of M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks into a logical sequence, classification, structure, or calculation.
- Connect M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks?
- How would you distinguish M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks from a closely related idea?
- Where would an engineer or technician encounter M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.

- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M1.04 Summarize Network attacks 4 Understanding Explain Counter measures for Network M1.05 2 Understanding attacks താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ച് കൃത്യമായ technical term ഉപയോഗിച്ച് നിങ്ങളുടെ definition നൽകുക. principle, named parts/steps, application, limitation എന്നിവയെക്കുറിച്ചും ഉൾപ്പെടുത്തുക. English terminology, symbols, units, diagram labels ഉപയോഗിച്ച് ഉത്തരം നൽകുക. Exam answer രീതിയിൽ concept-യെക്കുറിച്ച് ചുരുക്കത്തിൽ വിശദീകരിക്കുക.

– Official syllabus point 5: M1.05 2 Understanding attacks Contents

Exact source point

Syllabus text: M1.05 2 Understanding attacks Contents

How to understand and study it

This point is an official syllabus requirement: “M1.05 2 Understanding attacks Contents”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M1.05 2 Understanding attacks Contents ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M1.05 2 Understanding attacks Contents is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.05 2 Understanding attacks Contents using the official terminology retained above.
- Organise the important elements of M1.05 2 Understanding attacks Contents into a logical sequence, classification, structure, or calculation.
- Connect M1.05 2 Understanding attacks Contents with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.05 2 Understanding attacks Contents with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.05 2 Understanding attacks Contents. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.05 2 Understanding attacks Contents is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.05 2 Understanding attacks Contents, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.05 2 Understanding attacks Contents, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.05 2 Understanding attacks Contents?
- How would you distinguish M1.05 2 Understanding attacks Contents from a closely related idea?
- Where would an engineer or technician encounter M1.05 2 Understanding attacks Contents, and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.05 2 Understanding attacks Contents incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.05 2 Understanding attacks Contents താഴെ topic തിരഞ്ഞെടുക്കുന്നതിനായി ഉപയോഗിക്കുക exact technical term ഉപയോഗിക്കുക. definition ഉപയോഗിക്കുക principle, named parts/steps, application, limitation ഉപയോഗിക്കുക ഉപയോഗിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക ഉപയോഗിക്കുക. Exam answer ഉപയോഗിക്കുക concept-ഉപയോഗിക്കുക ഉപയോഗിക്കുക-ഉപയോഗിക്കുക ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

Learning goals

- Define hacking, ethical hacking, penetration testing, and related ethical boundaries.
- Classify malware and network/system attacks.
- Select conceptual countermeasures using confidentiality, integrity, and availability.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — Hacking and ethical hacking (2 hours)

Concept and explanation

Hacking is the use of technical methods to access, manipulate, or test systems; the intent and authorisation determine whether an activity is malicious, unauthorised, or ethical. Ethical hacking is a controlled security assessment performed within written scope, rules of engagement, and reporting requirements. Penetration testing simulates authorised attack paths to identify exploitable weaknesses and verify controls.

Malayalam support: Hacking തെക്നിക്കൽ മെത്തഡ് ഉപയോഗിച്ച് സിസ്റ്റം അക്സസ്, മാനിപുലേറ്റ്, അല്ലെങ്കിൽ ടെസ്റ്റ് ചെയ്യുക; ഉദ്ദേശ്യം അനുമതി തീരുമാനിക്കുന്നു; അനുമതിയും ഉദ്ദേശ്യവും ദുഷ്കൃത്യമായ/നിയമപരമായ ആക്ട് ആയിരിക്കും. നിയമപരമായ ഹാക്കിംഗ് എഴുതിയ അനുമതി, സ്കോപ്പ്, റൂൾസ് ഓഫ് എംഗേജ്മെന്റ് റിപ്പോർട്ടിംഗ് റിക്വയർമെന്റുകൾ ഉൾക്കൊള്ളുന്നു.

Study points

- Define ethical hacking and penetration testing; state the need for permission and scope.

Engineering / workplace application: Authorised penetration tests help organisations find weaknesses before criminals exploit them.

Illustrative example: A test plan may specify target IP ranges, testing dates, allowed tools, emergency contacts, and reporting format.

Common mistake: Scanning or exploiting a system without permission is not ethical hacking, even if the person claims educational intent.

Handbook treatment

M1.01 — Hacking and ethical hacking (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.01 — Hacking and ethical hacking (2 hours) using the official terminology retained above.
- Organise the important elements of M1.01 — Hacking and ethical hacking (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.01 — Hacking and ethical hacking (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.01 — Hacking and ethical hacking (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.01 — Hacking and ethical hacking (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.01 — Hacking and ethical hacking (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.01 — Hacking and ethical hacking (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.01 — Hacking and ethical hacking (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 — Hacking and ethical hacking (2 hours)?
- How would you distinguish M1.01 — Hacking and ethical hacking (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.01 — Hacking and ethical hacking (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 — Hacking and ethical hacking (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.01 — Hacking and ethical hacking (2 hours) താഴെ topic നൽകുന്നതിനുള്ള ഉത്തരം exact technical term നൽകുന്നതിനുള്ളതാണ്. താഴെ definition നൽകുന്നതിനുള്ള principle, named parts/steps, application, limitation നൽകുന്നതിനുള്ള ഉത്തരം. English terminology, symbols, units, diagram labels നൽകുന്നതിനുള്ള ഉത്തരം. Exam answer നൽകുന്നതിനുള്ള concept-നൽകുന്നതിനുള്ള ഉത്തരം-നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ളതാണ്.

– M1.02 — Ethical issues related to hacking (2 hours)

Concept and explanation

Ethical issues include consent, privacy, data minimisation, scope control, safe handling of credentials, non-disruption, responsible disclosure, evidence protection, and avoiding unauthorised persistence. A tester must stop when the scope is exceeded, protect discovered personal data, and report critical findings promptly. Laws and organisational policies apply even when a vulnerability is easy to exploit.

Malayalam support: Consent, privacy, scope, evidence protection, responsible disclosure എന്നിവയ്ക്ക് ethical hacking-ന് പരിധി വെക്കേണ്ടതാണ്. Permission നൽകാത്ത test നടത്തുന്നത് ethical-നോട് പൊരുത്തപ്പെടുന്നില്ല.

Study points

- List ethical duties before, during, and after a security test.

Engineering / workplace application: Security consultants use rules of engagement and confidentiality agreements to protect clients and users.

Illustrative example: If a test unexpectedly reaches a third-party system, the tester should stop, record the event, and notify the authorised contact.

Common mistake: Publishing live credentials or personal data as proof of a finding causes harm and violates responsible practice.

Handbook treatment

M1.02 — Ethical issues related to hacking (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M1.02 — Ethical issues related to hacking (2 hours) using the official terminology retained above.
- Organise the important elements of M1.02 — Ethical issues related to hacking (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.02 — Ethical issues related to hacking (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.02 — Ethical issues related to hacking (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.02 — Ethical issues related to hacking (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M1.02 — Ethical issues related to hacking (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M1.02 — Ethical issues related to hacking (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.02 — Ethical issues related to hacking (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 — Ethical issues related to hacking (2 hours)?
- How would you distinguish M1.02 — Ethical issues related to hacking (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.02 — Ethical issues related to hacking (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 — Ethical issues related to hacking (2 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M1.02 — Ethical issues related to hacking (2 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും exact technical term നൽകിയിരിക്കുന്നതെന്ന്. താഴെ definition
നൽകിയിരിക്കുന്ന principle, named parts/steps, application, limitation നൽകിയിരിക്കുന്നവയെ
നൽകിയിരിക്കുന്നു. English terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്ന
നൽകിയിരിക്കുന്നു. Exam answer നൽകിയിരിക്കുന്ന concept-നൽകിയിരിക്കുന്നവയെ നൽകിയിരിക്കുന്നു
നൽകിയിരിക്കുന്നു.

– M1.03 — Malware classes, actions and protection (4 hours)

Concept and explanation

Malware is malicious software designed to disrupt, damage, spy, extort, or gain unauthorised access. Viruses attach to files and spread when executed; worms self-propagate across networks; Trojans masquerade as legitimate software; spyware monitors activity; adware displays unwanted advertising and may track behaviour. Protection includes patching, least privilege, endpoint protection, secure backups, application control, user awareness, network segmentation, and monitoring.

Malayalam support: Virus file-കുറിച്ചു attach ചെയ്യുന്നു; worm network ൽ self-propagate ചെയ്യുന്നു; Trojan legitimate software ന്റെ രൂപം കൈമാറുന്നു; spyware monitor ചെയ്യുന്നു; adware unwanted ads/ tracking ചെയ്യുന്നു. Protection ൽ layers ഉണ്ട്.

Study points

- Classify the listed malware and state one protection measure for each.

Engineering / workplace application: Endpoint security, backups, patching, segmentation, and awareness reduce malware impact in organisations.

Illustrative example: A restore-tested offline backup can limit ransomware impact, while least privilege reduces the malware actions available to a compromised account.

Common mistake: Antivirus alone cannot compensate for unpatched systems, weak credentials, or unsafe backups.

Handbook treatment

M1.03 — Malware classes, actions and protection (4 hours) must be understood as a control-of-risk topic. Identify the hazard, the possible harm, the conditions that increase the risk, and the hierarchy of controls that prevents the incident.

Learning objectives

- Define and scope M1.03 — Malware classes, actions and protection (4 hours) using the official terminology retained above.
- Organise the important elements of M1.03 — Malware classes, actions and protection (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.03 — Malware classes, actions and protection (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.03 — Malware classes, actions and protection (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.03 — Malware classes, actions and protection (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the source of energy or unsafe condition.
- Describe the possible consequence and the people, equipment, or environment exposed.
- Apply elimination, substitution, engineering control, administrative control, and personal protective equipment in that order.
- State inspection, isolation, emergency response, reporting, and recovery requirements where relevant.

Engineering application lens

In engineering practice, M1.03 — Malware classes, actions and protection (4 hours) is part of normal design, operation, maintenance, and troubleshooting—not a separate final paragraph. The safest answer connects the control measure to the hazard and explains why the measure is effective.

Worked answer method

Given: the work activity, equipment, hazard, or incident condition.

Required: identify risk and propose controls.

Method: describe hazard → consequence → control → verification → emergency action.

Answer: give specific controls, responsible action, and one condition that must be checked before work begins.

Exam answer blueprint

For a short-answer question on M1.03 — Malware classes, actions and protection (4 hours), begin with the definition or principle and include the decisive feature.

For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.03 — Malware classes, actions and protection (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 — Malware classes, actions and protection (4 hours)?
- How would you distinguish M1.03 — Malware classes, actions and protection (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.03 — Malware classes, actions and protection (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 — Malware classes, actions and protection (4 hours) incorrect?

Common mistakes and safety emphasis

- Naming PPE as the only control when isolation or guarding is required.
- Describing a hazard without stating the consequence.
- Ignoring stored energy, restart, access, housekeeping, or emergency isolation.
- Treating a safety instruction as optional or disconnected from the operating procedure.

Malayalam support: M1.03 — Malware classes, actions and protection (4 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels Exam answer concept- -

– M1.04 — Network attacks (4 hours)

Concept and explanation

The syllabus includes DoS, DDoS, buffer overflow, ping of death, session hijacking, brute-force, man-in-the-middle, dictionary, and replay attacks. DoS exhausts a service resource; DDoS distributes the traffic or workload across many sources. A buffer overflow abuses inadequate bounds checking. Session hijacking steals or reuses session state. Brute-force and dictionary attacks try credentials; MITM intercepts or alters communication; replay reuses captured valid messages.

Malayalam support: DoS service unavailable റിസോഴ്സ് resource exhaust റിസോഴ്സ്; DDoS many sources വിതരണം. Buffer overflow bounds-checking weakness തകരാറുണ്ട്. Session hijacking, MITM, brute force, dictionary, replay കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം, കൈമാറ്റം.

Study points

- Map each attack to a target property and one countermeasure.

Engineering / workplace application: Network monitoring, rate limiting, secure protocols, input validation, MFA, and session protection are practical controls.

Illustrative example: TLS protects communication confidentiality/integrity, but authentication and endpoint security are still needed to reduce MITM and session abuse.

Common mistake: Do not describe an attack only by its name; explain what resource, message, session, or input it targets.

Handbook treatment

M1.04 — Network attacks (4 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M1.04 — Network attacks (4 hours) using the official terminology retained above.
- Organise the important elements of M1.04 — Network attacks (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.04 — Network attacks (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.04 — Network attacks (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.04 — Network attacks (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M1.04 — Network attacks (4 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M1.04 — Network attacks (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.04 — Network attacks (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 — Network attacks (4 hours)?
- How would you distinguish M1.04 — Network attacks (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.04 — Network attacks (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 — Network attacks (4 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M1.04 — Network attacks (4 hours) താഴെ topic

തയ്യാറാക്കേണ്ടതാണ് താഴെ exact technical term തയ്യാറാക്കേണ്ടതാണ്. താഴെ definition

തയ്യാറാക്കേണ്ടതാണ് principle, named parts/steps, application, limitation തയ്യാറാക്കേണ്ടതാണ്

തയ്യാറാക്കേണ്ടതാണ്. English terminology, symbols, units, diagram labels തയ്യാറാക്കേണ്ടതാണ്

തയ്യാറാക്കേണ്ടതാണ്. Exam answer തയ്യാറാക്കേണ്ടതാണ് concept-തയ്യാറാക്കേണ്ടതാണ്-തയ്യാറാക്കേണ്ടതാണ്

തയ്യാറാക്കേണ്ടതാണ്.

Concept and explanation

Countermeasures are layered: secure configuration and patching, firewalls and segmentation, strong authentication and MFA, encrypted protocols, input validation, rate limiting, anti-DDoS services, secure session management, logging, alerting, backups, and incident response. The CIA triad helps classify the impact: confidentiality prevents unauthorised disclosure, integrity prevents unauthorised change, and availability keeps services usable.

Malayalam support: CIA triad- confidentiality, integrity, availability. Firewall, patching, MFA, encryption, validation, rate limiting, monitoring, backup layered defence.

Study points

- Relate a countermeasure to confidentiality, integrity, or availability.

Engineering / workplace application: Defence-in-depth prevents one failed control from exposing the whole service.

Illustrative example: A service can combine MFA, rate limits, account lockout policy, secure logging, and incident response to reduce brute-force impact.

Common mistake: A single firewall rule is not a complete countermeasure for credential theft, insider risk, or application vulnerabilities.

Handbook treatment

M1.05 — Countermeasures for network attacks (2 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M1.05 — Countermeasures for network attacks (2 hours) using the official terminology retained above.
- Organise the important elements of M1.05 — Countermeasures for network attacks (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.05 — Countermeasures for network attacks (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Hacking, Malware, Network Attacks and Countermeasures.
- Write an examination answer on M1.05 — Countermeasures for network attacks (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M1.05 — Countermeasures for network attacks (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M1.05 — Countermeasures for network attacks (2 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M1.05 — Countermeasures for network attacks (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M1.05 — Countermeasures for network attacks (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.05 — Countermeasures for network attacks (2 hours)?
- How would you distinguish M1.05 — Countermeasures for network attacks (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.05 — Countermeasures for network attacks (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.05 — Countermeasures for network attacks (2 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M1.05 — Countermeasures for network attacks (2 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels concept- diagram - answer concept-diagram .

Module summary: Ethical practice begins with authorisation and continues through layered controls, safe testing, and responsible reporting.

OFFICIAL MODULE 2

Footprinting, Social Engineering and Port Scanning

This module covers information gathering and scanning concepts. All demonstrations must use a permitted lab target and must not disrupt real systems.

Hours: 16

CO2 · Bloom level: Understanding / Applying

Handbook study routine: Complete Module 2 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 2

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding
M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding
M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying
M2.04 6 Applying tools Series Test - I 1

Point-by-point study guide

– **Official syllabus point 1: M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding**

Exact source point

Syllabus text: M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding

How to understand and study it

This point is an official syllabus requirement: “M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.01 Demonstrate foot printing, tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding using the official terminology retained above.
- Organise the important elements of M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer

verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding?
- How would you distinguish M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding from a closely related idea?
- Where would an engineer or technician encounter M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 Demonstrate foot printing and tools used for 3 Applying it M2.02 Explain competitive intelligence methods 3 Understanding incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.01 Demonstrate foot printing and tools used for
3 Applying it M2.02 Explain competitive intelligence methods 3
Understanding താഴെ പറയുന്ന വിഷയം പരിശോധിക്കുക. താഴെ പറയുന്ന exact technical term
പരിശോധിക്കുക. താഴെ പറയുന്ന definition പരിശോധിക്കുക principle, named parts/steps,
application, limitation പരിശോധിക്കുക. English terminology,
symbols, units, diagram labels പരിശോധിക്കുക. Exam answer
പരിശോധിക്കുക concept-താഴെ പറയുന്ന വിഷയം പരിശോധിക്കുക.

– **Official syllabus point 2: M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding**

Exact source point

Syllabus text: M2.02 Explain competitive intelligence methods 3 Understanding
M2.03 Illustrate Social Engineering Methods 4 Understanding

How to understand and study it

This point is an official syllabus requirement: “M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding using the official terminology retained above.
- Organise the important elements of M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level

answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding?
- How would you distinguish M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding from a closely related idea?
- Where would an engineer or technician encounter M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 Explain competitive intelligence methods 3 Understanding M2.03 Illustrate Social Engineering Methods 4 Understanding incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M2.02 Explain competitive intelligence methods 3

Understanding M2.03 Illustrate Social Engineering Methods 4

Understanding താഴെ topic നൽകുന്നതിനായി ഉപയോഗിക്കേണ്ട exact technical term

ഉപയോഗിക്കേണ്ടതാണ്. ഉപയോഗിക്കേണ്ട definition നൽകുന്നതിനായി principle, named parts/steps, application, limitation നൽകേണ്ടതാണ് ഉപയോഗിക്കേണ്ടതാണ്. English terminology, symbols, units, diagram labels നൽകേണ്ടതാണ് ഉപയോഗിക്കേണ്ടതാണ്. Exam answer നൽകേണ്ടതാണ് concept-നൽകേണ്ടതാണ് ഉപയോഗിക്കേണ്ടതാണ്.

– **Official syllabus point 3: M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying**

Exact source point

Syllabus text: M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying

How to understand and study it

This point is an official syllabus requirement: “M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods, M2.04 6 Applying ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying using the official terminology retained above.
- Organise the important elements of M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying into a logical sequence, classification, structure, or calculation.
- Connect M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying. It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying?
- How would you distinguish M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying from a closely related idea?
- Where would an engineer or technician encounter M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 Illustrate Social Engineering Methods 4 Understanding Demonstrate port scanning, methods and M2.04 6 Applying incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.

- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M2.03 Illustrate Social Engineering Methods 4

Understanding Demonstrate port scanning, methods and M2.04 6 Applying
 താഴെ topic നൽകിയിരിക്കുന്നത് നൽകി exact technical term നൽകിയിരിക്കുന്നു. താഴെ
 definition നൽകിയിരിക്കുന്നു principle, named parts/steps, application, limitation
 നൽകിയിരിക്കുന്നു. English terminology, symbols, units, diagram
 labels നൽകിയിരിക്കുന്നു. Exam answer നൽകിയിരിക്കുന്നു concept-നൽകി നൽകി-
 നൽകി നൽകി നൽകിയിരിക്കുന്നു.

– Official syllabus point 4: M2.04 6 Applying tools Series Test – I 1

Exact source point

Syllabus text: M2.04 6 Applying tools Series Test – I 1

How to understand and study it

This point is an official syllabus requirement: “M2.04 6 Applying tools Series Test – I 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M2.04 6 Applying tools Series Test – I 1 ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M2.04 6 Applying tools Series Test – I 1 is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.04 6 Applying tools Series Test – I 1 using the official terminology retained above.
- Organise the important elements of M2.04 6 Applying tools Series Test – I 1 into a logical sequence, classification, structure, or calculation.
- Connect M2.04 6 Applying tools Series Test – I 1 with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.04 6 Applying tools Series Test – I 1 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.04 6 Applying tools Series Test – I 1. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.04 6 Applying tools Series Test – I 1 is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.04 6 Applying tools Series Test – I 1, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.04 6 Applying tools Series Test – I 1, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 6 Applying tools Series Test – I 1?
- How would you distinguish M2.04 6 Applying tools Series Test – I 1 from a closely related idea?
- Where would an engineer or technician encounter M2.04 6 Applying tools Series Test – I 1, and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 6 Applying tools Series Test – I 1 incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

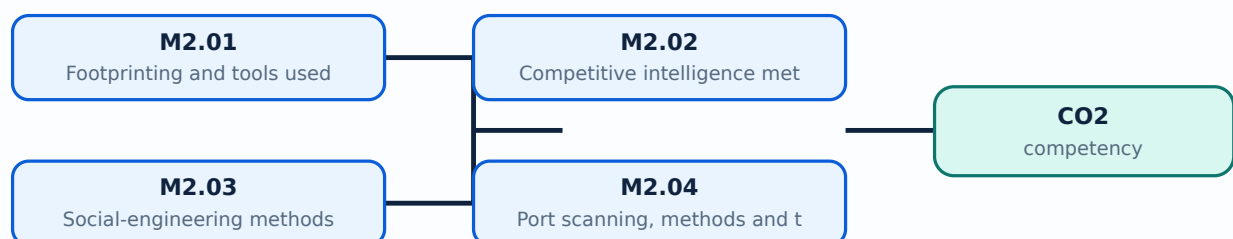
Malayalam support: M2.04 6 Applying tools Series Test – I 1 താഴെ പറയുന്ന വിഷയം കൃത്യമായ technical term ഉപയോഗിച്ച് വിശദീകരിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾപ്പെടെ വിശദീകരിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിച്ച് വിശദീകരിക്കുക. Exam answer concept-യെക്കുറിച്ച് വിശദീകരിക്കുക.

Learning goals

- Explain footprinting, competitive intelligence, DNS zone transfer, and social-engineering methods.
- Describe port-scan types and the purpose of tools named in the syllabus.
- Interpret scanning results without treating them as proof of exploitability.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — Footprinting and tools used for it (3 hours)

Concept and explanation

Footprinting is the systematic collection of information about an organisation, domain, network, technology stack, and public exposure before a security assessment. Passive methods use public records and observable information; active methods query or interact with systems and require explicit authorisation. Web search, domain records, certificate information, DNS data, and approved assessment tools can reveal attack surface. Findings should be validated and documented with source and time.

Malayalam support: Footprinting target-പ്രദേശം public information, domain, network, technology, exposure പരസ്യമായ വിവരങ്ങൾ ശേഖരിക്കുക. Passive collection പരസ്യമായ വിവരങ്ങൾ ശേഖരിക്കുക; active query permission-പ്രദേശം ശേഖരിക്കുക.

Study points

- Differentiate passive and active footprinting.
- Record source, timestamp, confidence, and scope for collected information.
- Use only authorised tools and targets.

Engineering / workplace application: Asset inventory and attack-surface management use the same information-gathering discipline for defence.

Illustrative example: A tester may collect a domain certificate record passively, then request permission before running active DNS or service queries.

Common mistake: Publicly available information is not automatically safe to probe or republish.

Handbook treatment

M2.01 — Footprinting and tools used for it (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.01 — Footprinting and tools used for it (3 hours) using the official terminology retained above.
- Organise the important elements of M2.01 — Footprinting and tools used for it (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.01 — Footprinting and tools used for it (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.01 — Footprinting and tools used for it (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.01 — Footprinting and tools used for it (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.01 — Footprinting and tools used for it (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.01 — Footprinting and tools used for it (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.01 — Footprinting and tools used for it (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 — Footprinting and tools used for it (3 hours)?
- How would you distinguish M2.01 — Footprinting and tools used for it (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.01 — Footprinting and tools used for it (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 — Footprinting and tools used for it (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.01 — Footprinting and tools used for it (3 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും exact technical term നൽകിയിരിക്കുന്നു. താഴെ definition
നൽകിയിരിക്കുന്ന principle, named parts/steps, application, limitation നൽകിയിരിക്കുന്നവയിൽ
ഏതെങ്കിലും. English terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്നു.
Exam answer നൽകിയിരിക്കുന്ന concept-നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും
നൽകിയിരിക്കുന്നു.

Concept and explanation

Competitive intelligence in security assessment means structured analysis of publicly available information about technologies, suppliers, domains, job postings, exposed services, and organisational change. The information should be collected lawfully, evaluated for reliability, and used to understand likely technology and risk rather than to target individuals. Corroboration is important because public data may be stale or misleading.

Malayalam support: Competitive intelligence public and lawful information structured അനാലൈസ് ചെയ്യപ്പെടുന്നു. Source reliability, timestamp, corroboration പരിശോധിക്കപ്പെടുന്നു; individuals-ക്ക് target ചെയ്യപ്പെടുന്നു.

Study points

- Explain lawful collection, validation, and analysis.
- Separate fact from inference when reporting intelligence.
- Avoid collecting unnecessary personal information.

Engineering / workplace application: Defenders use intelligence to prioritise patching, exposure reduction, and supplier risk review.

Illustrative example: Two independent sources showing the same technology version provide stronger evidence than one unverified web page.

Common mistake: Treating a job advertisement or old technology reference as proof of current deployment can produce a false conclusion.

Handbook treatment

M2.02 — Competitive intelligence methods (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.02 — Competitive intelligence methods (3 hours) using the official terminology retained above.
- Organise the important elements of M2.02 — Competitive intelligence methods (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.02 — Competitive intelligence methods (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.02 — Competitive intelligence methods (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.02 — Competitive intelligence methods (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.02 — Competitive intelligence methods (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.02 — Competitive intelligence methods (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.02 — Competitive intelligence methods (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 — Competitive intelligence methods (3 hours)?
- How would you distinguish M2.02 — Competitive intelligence methods (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.02 — Competitive intelligence methods (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 — Competitive intelligence methods (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.02 — Competitive intelligence methods (3 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും exact technical term നൽകിയിരിക്കുന്നു. താഴെ definition
നൽകിയിരിക്കുന്നവയിൽ principle, named parts/steps, application, limitation നൽകിയിരിക്കുന്നു. താഴെ
നൽകിയിരിക്കുന്നു. English terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്നു.
നൽകിയിരിക്കുന്നു. Exam answer നൽകിയിരിക്കുന്നു concept-നൽകിയിരിക്കുന്നു-നൽകിയിരിക്കുന്നു
നൽകിയിരിക്കുന്നു.

Concept and explanation

Social engineering manipulates people rather than software. Shoulder surfing observes information directly; dumpster diving searches discarded material; piggybacking follows an authorised person into a restricted area; phishing uses deceptive messages or pages to obtain action or information. Awareness, verification, physical access control, secure disposal, MFA, reporting channels, and least privilege reduce risk. Ethical simulations need consent, safety boundaries, and a clear learning objective.

Malayalam support: Shoulder surfing, dumpster diving, piggybacking, phishing ഹ്യൂമൻ ബഹുവൃത്തിയെ ഹിസ്സാസ്സാക്കുവാൻ ഉപയോഗിക്കുന്ന രീതികൾ. Awareness, physical security, secure disposal, MFA, reporting ഹിസ്സാസ്സാക്കുവാൻ ഉപയോഗിക്കുന്ന കൗണ്ടർമേജറുകൾ.

Study points

- Define the listed methods and one prevention for each.
- Explain why awareness training must be paired with technical and physical controls.
- Keep simulations authorised and non-harmful.

Engineering / workplace application: Security awareness programmes use controlled phishing simulations and physical-security exercises to improve reporting behaviour.

Illustrative example: A verified callback procedure can stop a fraudulent password-reset request even when the message looks plausible.

Common mistake: Blaming a user after an incident without fixing weak process, authentication, or access control does not provide a complete defence.

Handbook treatment

M2.03 — Social-engineering methods (4 hours) is best studied as an engineering system rather than as an isolated name. Relate the construction of each main part to its function, then follow the energy or material flow from input to output.

Learning objectives

- Define and scope M2.03 — Social-engineering methods (4 hours) using the official terminology retained above.
- Organise the important elements of M2.03 — Social-engineering methods (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.03 — Social-engineering methods (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.03 — Social-engineering methods (4 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.03 — Social-engineering methods (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the purpose, input, output, and operating conditions.
- Draw or imagine the main construction and label the components that determine performance.
- Explain the operating sequence using cause-and-effect statements.
- Connect performance, losses, limitations, maintenance, and application to the operating principle.

Engineering application lens

In practice, M2.03 — Social-engineering methods (4 hours) is selected by matching the required duty with capacity, efficiency, controllability, reliability, safety, maintenance needs, and environmental conditions. A good diploma-level answer explains not only how it works, but why that construction is suitable for the stated application.

Worked answer method

Given: the equipment type, duty, operating condition, or fault described in the question.

Required: construction, working, selection, performance, or diagnosis as requested.

Method: begin with a labelled block or component list; explain the energy/material path; relate each observation to a likely cause or operating principle.

Answer: finish with the application, limitation, and one relevant safety or maintenance point.

Exam answer blueprint

For a short-answer question on M2.03 — Social-engineering methods (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.03 — Social-engineering methods (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 — Social-engineering methods (4 hours)?
- How would you distinguish M2.03 — Social-engineering methods (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.03 — Social-engineering methods (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 — Social-engineering methods (4 hours) incorrect?

Common mistakes and safety emphasis

- Listing parts without stating their functions.
- Describing operation without identifying the input and output.
- Mixing the construction of similar equipment types.
- Ignoring ratings, operating limits, guarding, lubrication, isolation, or maintenance conditions.

Malayalam support: M2.03 — Social-engineering methods (4 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾപ്പെടെയുള്ള വിവരങ്ങൾ നൽകുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer രേഖപ്പെടുത്തുക concept-ബോക്സ് ഉപയോഗിച്ച് ഉത്തരം നൽകുക.

Concept and explanation

Port scanning probes network endpoints to identify reachable ports and infer services. TCP connect scans complete a connection; SYN scans examine handshake responses without completing a full connection in the usual case; UDP scans require different interpretation because responses may be absent or filtered. Ping sweeps identify responding hosts. The syllabus names Nmap and Nessus as tools and also includes crafting IP packets and spoofing. Tool output is an observation that must be validated, documented, and handled within scope.

Malayalam support: Port scan reachable ports/services ഏതെങ്കിലും ഏതെങ്കിലും. TCP connect, SYN, UDP scan, ping sweep ഏതെങ്കിലും response pattern ഏതെങ്കിലും. Nmap/Nessus ഏതെങ്കിലും tools authorised lab-ൽ ഏതെങ്കിലും ഏതെങ്കിലും.

Study points

- Explain what a port state can and cannot prove.
- Differentiate host discovery, port scanning, and vulnerability assessment.
- State that packet crafting and spoofing require a controlled lab and explicit permission.

Engineering / workplace application: Administrators use approved scanning to find exposed services, remove unnecessary ports, and verify firewall rules.

Illustrative example: A safe lab scan of a student-owned virtual network can identify an intended test web service, after which the service version is validated separately.

Common mistake: An open port is not proof of a vulnerability, and a filtered or silent port is not proof that no service exists.

Handbook treatment

M2.04 — Port scanning, methods and tools (6 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M2.04 — Port scanning, methods and tools (6 hours) using the official terminology retained above.
- Organise the important elements of M2.04 — Port scanning, methods and tools (6 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.04 — Port scanning, methods and tools (6 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Footprinting, Social Engineering and Port Scanning.
- Write an examination answer on M2.04 — Port scanning, methods and tools (6 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M2.04 — Port scanning, methods and tools (6 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M2.04 — Port scanning, methods and tools (6 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M2.04 — Port scanning, methods and tools (6 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M2.04 — Port scanning, methods and tools (6 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 — Port scanning, methods and tools (6 hours)?
- How would you distinguish M2.04 — Port scanning, methods and tools (6 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.04 — Port scanning, methods and tools (6 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 — Port scanning, methods and tools (6 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M2.04 — Port scanning, methods and tools (6 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയ്ക്ക് താഴെ exact technical term നൽകിയിരിക്കുന്നു. താഴെ definition
നൽകിയിരിക്കുന്നവയ്ക്ക് principle, named parts/steps, application, limitation നൽകിയിരിക്കുന്നു
നൽകിയിരിക്കുന്നു. English terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്നു
നൽകിയിരിക്കുന്നു. Exam answer നൽകിയിരിക്കുന്നവയ്ക്ക് concept-നൽകിയിരിക്കുന്നു-നൽകിയിരിക്കുന്നു
നൽകിയിരിക്കുന്നു.

Module summary: Reconnaissance is evidence gathering, not exploitation: scope, safety, validation, and documentation control every demonstration.

OFFICIAL MODULE 3

Operating-System Vulnerabilities and Hardening

This module compares Windows and Linux vulnerability areas and emphasises hardening, identification, authentication, and countermeasures.

Hours: 15

CO3 • Bloom level: Understanding

Handbook study routine: Complete Module 3 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 3

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

[View extracted official source transcript](#)

M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks M3.04 3 Understanding attacks Content:

Point-by-point study guide

– **Official syllabus point 1: M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding**

Exact source point

Syllabus text: M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding

How to understand and study it

This point is an official syllabus requirement: “M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding using the official terminology retained above.
- Organise the important elements of M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding into a logical sequence, classification, structure, or calculation.
- Connect M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer

verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding?
- How would you distinguish M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding from a closely related idea?
- Where would an engineer or technician encounter M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 Summarize windows Vulnerable components 6 Understanding M3.02 Explain methods for hardening windows 3 Understanding incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.01 Summarize windows Vulnerable components
6 Understanding M3.02 Explain methods for hardening windows 3
Understanding താഴെ topic നൽകുന്നതിനുള്ള ഉത്തരം exact technical term
നൽകുന്നതിനുള്ള. ഉത്തരം definition നൽകുന്നതിനുള്ള principle, named parts/steps,
application, limitation നൽകുന്നതിനുള്ള ഉത്തരം. English terminology,
symbols, units, diagram labels നൽകുന്നതിനുള്ള ഉത്തരം. Exam answer
നൽകുന്നതിനുള്ള concept-നൽകുന്നതിനുള്ള ഉത്തരം-നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള.

Exact source point

Syllabus text: M3.02 Explain methods for hardening windows 3 Understanding
M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter
measures against Linux

How to understand and study it

This point is an official syllabus requirement: “M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux”. Record the relevant quantity, symbol, unit, relation or formula, and the interpretation of the result. Do not memorise a number without its unit and condition. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Record the symbol, SI unit, governing relation, and one correctly labelled calculation.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: syllabus point ഉള്ളതുകൊണ്ട് M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux ഉള്ള technical terms English-ഉള്ളതുകൊണ്ട് ഉള്ളതുകൊണ്ട്. ഉള്ളതുകൊണ്ട് definition ഉള്ളതുകൊണ്ട് principle ഉള്ളതുകൊണ്ട്; ഉള്ളതുകൊണ്ട് term-ഉള്ളതുകൊണ്ട് function, difference, application ഉള്ളതുകൊണ്ട് ഉള്ളതുകൊണ്ട് ഉള്ളതുകൊണ്ട്. Diagram, sequence, formula, unit ഉള്ളതുകൊണ്ട് ഉള്ളതുകൊണ്ട് ഉള്ളതുകൊണ്ട് revision ഉള്ളതുകൊണ്ട്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux using the official terminology retained above.
- Organise the important elements of M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux into a logical sequence, classification, structure, or calculation.
- Connect M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux?
- How would you distinguish M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux from a closely related idea?
- Where would an engineer or technician encounter M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS

vulnerabilities 3 Understanding Describe counter measures against Linux, and what must be checked before using it?

- What common mistake would make an answer or operation involving M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.02 Explain methods for hardening windows 3 Understanding M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ച് കൃത്യമായ technical term ഉപയോഗിക്കുക. താഴെ പറയുന്ന definition ഉപയോഗിക്കുക principle, named parts/steps, application, limitation ഉപയോഗിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer ഉപയോഗിക്കുക concept-ഉപയോഗിക്കുക ഉപയോഗിക്കുക-ഉപയോഗിക്കുക ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

– **Official syllabus point 3: M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks**

Exact source point

Syllabus text: M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks

How to understand and study it

This point is an official syllabus requirement: “M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks”. Record the relevant quantity, symbol, unit, relation or formula, and the interpretation of the result. Do not memorise a number without its unit and condition. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Record the symbol, SI unit, governing relation, and one correctly labelled calculation.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks using the official terminology retained above.
- Organise the important elements of M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks into a logical sequence, classification, structure, or calculation.
- Connect M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or

designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks?
- How would you distinguish M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks from a closely related idea?
- Where would an engineer or technician encounter M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 Summarize Linux OS vulnerabilities 3 Understanding Describe counter measures against Linux M3.04 3 Understanding attacks incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.03 Summarize Linux OS vulnerabilities 3

Understanding Describe counter measures against Linux M3.04 3

Understanding attacks താഴെ topic നൽകുന്നതിനുള്ള ഉത്തരം exact technical term നൽകുന്നതിനുള്ള. താഴെ definition നൽകുന്നതിനുള്ള principle, named parts/steps, application, limitation നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള. English terminology, symbols, units, diagram labels നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള. Exam answer നൽകുന്നതിനുള്ള concept-നൽകുന്നതിനുള്ള ഉത്തരം-നൽകുന്നതിനുള്ള ഉത്തരം നൽകുന്നതിനുള്ള.

– Official syllabus point 4: M3.04 3 Understanding attacks Content

Exact source point

Syllabus text: M3.04 3 Understanding attacks Content

How to understand and study it

This point is an official syllabus requirement: “M3.04 3 Understanding attacks Content”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M3.04 3 Understanding attacks Content ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ്. ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M3.04 3 Understanding attacks Content is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.04 3 Understanding attacks Content using the official terminology retained above.
- Organise the important elements of M3.04 3 Understanding attacks Content into a logical sequence, classification, structure, or calculation.
- Connect M3.04 3 Understanding attacks Content with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.04 3 Understanding attacks Content with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 3 Understanding attacks Content. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.04 3 Understanding attacks Content is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.04 3 Understanding attacks Content, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 3 Understanding attacks Content, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 3 Understanding attacks Content?
- How would you distinguish M3.04 3 Understanding attacks Content from a closely related idea?
- Where would an engineer or technician encounter M3.04 3 Understanding attacks Content, and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 3 Understanding attacks Content incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.04 3 Understanding attacks Content താഴെ topic തിരഞ്ഞെടുക്കുന്നതിന് ഉപയോഗിക്കുക exact technical term ഉപയോഗിക്കുക. definition ഉപയോഗിക്കുക principle, named parts/steps, application, limitation ഉപയോഗിക്കുക ഉപയോഗിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക ഉപയോഗിക്കുക. Exam answer ഉപയോഗിക്കുക concept-ഉപയോഗിക്കുക ഉപയോഗിക്കുക-ഉപയോഗിക്കുക ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

Learning goals

- Identify Windows vulnerable components listed in the syllabus.
- Explain Windows hardening and authentication controls.
- Recognise Linux vulnerability categories and corresponding countermeasures.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — Windows vulnerable components (6 hours)

Concept and explanation

The syllabus lists Windows file systems, Windows RPC, NetBIOS, Server Message Block, Common Internet File System, null sessions, web services, passwords, and authentication. Vulnerabilities may arise from legacy protocols, excessive exposure, weak permissions, unpatched services, insecure shares, or weak credentials. Assessment should identify the component, exposure, evidence, impact, and safe remediation rather than merely naming a tool.

Malayalam support: Windows file system, RPC, NetBIOS, SMB/CIFS, null session, web service, password/authentication ഏകദേശം exposure vulnerabilities ഏകദേശം. Component, evidence, impact, remediation ഏകദേശം report ഏകദേശം.

Study points

- Explain why legacy services and weak authentication increase attack surface.
- Relate file/share permissions to confidentiality and integrity.
- Document evidence without exposing real credentials.

Engineering / workplace application: Windows hardening and vulnerability management are routine enterprise administration activities.

Illustrative example: A security review can identify an unnecessary legacy file-sharing protocol, restrict it by network segment, patch the host, and verify the result.

Common mistake: Disabling a service without checking dependencies may break a business function; remediation must be tested.

Handbook treatment

M3.01 — Windows vulnerable components (6 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.01 — Windows vulnerable components (6 hours) using the official terminology retained above.
- Organise the important elements of M3.01 — Windows vulnerable components (6 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.01 — Windows vulnerable components (6 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.01 — Windows vulnerable components (6 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.01 — Windows vulnerable components (6 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.01 — Windows vulnerable components (6 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.01 — Windows vulnerable components (6 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.01 — Windows vulnerable components (6 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 — Windows vulnerable components (6 hours)?
- How would you distinguish M3.01 — Windows vulnerable components (6 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.01 — Windows vulnerable components (6 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 — Windows vulnerable components (6 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.01 — Windows vulnerable components (6 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും exact technical term ഉപയോഗിച്ച് definition
പ്രദാനം ചെയ്യുക principle, named parts/steps, application, limitation എന്നിവയിൽ ഏതെങ്കിലും
ഒന്നിനെക്കുറിച്ച്. English terminology, symbols, units, diagram labels ഉപയോഗിച്ച്
പ്രദാനം ചെയ്യുക. Exam answer നൽകുന്നതിനായി concept-യെക്കുറിച്ച് താഴെ നൽകിയിരിക്കുന്ന
രീതിയിൽ പ്രദാനം ചെയ്യുക.

Concept and explanation

Windows hardening includes timely patching, removal or restriction of unnecessary services, secure configuration of file sharing and RPC, strong password and MFA policy, least privilege, host firewall, endpoint protection, logging, secure remote administration, backup, and vulnerability verification. Hardening is a baseline-and-monitor process, not a one-time switch. Changes should be documented and tested.

Malayalam support: Windows hardening patching, unnecessary service restriction, strong authentication, least privilege, firewall, endpoint protection, logging, backup ഏറ്റെടുക്കുക. Baseline and continuous monitoring ഏറ്റെടുക്കുക.

Study points

- List preventive, detective, and recovery controls.
- Verify that hardening does not break required services.
- Use least privilege and secure administration.

Engineering / workplace application: System administrators apply hardening baselines before a server enters production.

Illustrative example: After restricting a legacy service, an administrator verifies required application connectivity, reviews logs, and records the approved baseline.

Common mistake: A hardening checklist without later monitoring allows configuration drift and new vulnerabilities.

Handbook treatment

M3.02 — Methods for hardening Windows (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.02 — Methods for hardening Windows (3 hours) using the official terminology retained above.
- Organise the important elements of M3.02 — Methods for hardening Windows (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.02 — Methods for hardening Windows (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.02 — Methods for hardening Windows (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.02 — Methods for hardening Windows (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.02 — Methods for hardening Windows (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.02 — Methods for hardening Windows (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.02 — Methods for hardening Windows (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 — Methods for hardening Windows (3 hours)?
- How would you distinguish M3.02 — Methods for hardening Windows (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.02 — Methods for hardening Windows (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 — Methods for hardening Windows (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.02 — Methods for hardening Windows (3 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും exact technical term നൽകിയിരിക്കുന്നു. താഴെ definition
നൽകിയിരിക്കുന്ന principle, named parts/steps, application, limitation നൽകിയിരിക്കുന്ന
നൽകിയിരിക്കുന്നു. English terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്നു
നൽകിയിരിക്കുന്നു. Exam answer നൽകിയിരിക്കുന്ന concept-നൽകിയിരിക്കുന്ന നൽകിയിരിക്കുന്ന
നൽകിയിരിക്കുന്നു.

Concept and explanation

Linux vulnerabilities can involve unpatched kernels or packages, weak file permissions, insecure services, exposed management interfaces, unsafe scripts, weak authentication, misconfigured sudo, and vulnerable web or network daemons. Identification uses configuration review, package and service inventories, logs, approved scanners, and manual verification. The operating system name alone does not determine security; configuration and maintenance matter.

Malayalam support: Linux vulnerability package updates, file permission, service exposure, sudo, authentication, scripts, web/network daemon configuration ഐഐഐഐഐഐഐഐഐഐ. OS name ഐഐഐഐ security ഐഐഐഐഐഐഐഐഐഐ.

Study points

- Categorise vulnerabilities as software, configuration, access, or service exposure.
- Use logs and configuration review together with approved tools.
- Avoid changing production systems during unauthorised testing.

Engineering / workplace application: Linux servers are hardened in web hosting, cloud, network appliances, and embedded environments.

Illustrative example: A review may find a world-writable configuration file, confirm its service impact, correct ownership and permissions, and monitor the service.

Common mistake: Running a scanner without understanding the host and validating findings leads to false positives or unsafe changes.

Handbook treatment

M3.03 — Linux OS vulnerabilities (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.03 — Linux OS vulnerabilities (3 hours) using the official terminology retained above.
- Organise the important elements of M3.03 — Linux OS vulnerabilities (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.03 — Linux OS vulnerabilities (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.03 — Linux OS vulnerabilities (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.03 — Linux OS vulnerabilities (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.03 — Linux OS vulnerabilities (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.03 — Linux OS vulnerabilities (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.03 — Linux OS vulnerabilities (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 — Linux OS vulnerabilities (3 hours)?
- How would you distinguish M3.03 — Linux OS vulnerabilities (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.03 — Linux OS vulnerabilities (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 — Linux OS vulnerabilities (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.03 — Linux OS vulnerabilities (3 hours) താഴെ topic

തയ്യാറാക്കേണ്ടതാണ് താഴെ exact technical term തയ്യാറാക്കേണ്ടതാണ്. താഴെ definition

തയ്യാറാക്കേണ്ടതാണ് principle, named parts/steps, application, limitation തയ്യാറാക്കേണ്ടതാണ്

തയ്യാറാക്കേണ്ടതാണ്. English terminology, symbols, units, diagram labels തയ്യാറാക്കേണ്ടതാണ്

തയ്യാറാക്കേണ്ടതാണ്. Exam answer തയ്യാറാക്കേണ്ടതാണ് concept-തയ്യാറാക്കേണ്ടതാണ്-തയ്യാറാക്കേണ്ടതാണ്

തയ്യാറാക്കേണ്ടതാണ്.

Concept and explanation

Linux countermeasures include patch management, minimal services, secure SSH configuration, key-based authentication, least privilege, firewalling, file-integrity monitoring, mandatory access controls where appropriate, backups, logging, alerting, and tested incident response. Administrators should separate service accounts, protect secrets, and review sudo rules. Countermeasures must be matched to the threat and operational requirement.

Malayalam support: Linux attacks-പാatching, minimal services, SSH hardening, key authentication, least privilege, firewall, integrity monitoring, access control, backup, logging പാatching പാatching.

Study points

- Give a layered Linux hardening sequence.
- Relate SSH, sudo, file permissions, and logging to attack reduction.
- Include recovery and monitoring, not only prevention.

Engineering / workplace application: Secure Linux administration supports servers, containers, network devices, and forensic workstations.

Illustrative example: A hardened SSH service can restrict users, use keys, limit network access, log attempts, and retain a tested recovery path.

Common mistake: Disabling password login without testing key recovery or console access can lock out administrators.

Handbook treatment

M3.04 — Countermeasures against Linux attacks (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M3.04 — Countermeasures against Linux attacks (3 hours) using the official terminology retained above.
- Organise the important elements of M3.04 — Countermeasures against Linux attacks (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.04 — Countermeasures against Linux attacks (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Operating-System Vulnerabilities and Hardening.
- Write an examination answer on M3.04 — Countermeasures against Linux attacks (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M3.04 — Countermeasures against Linux attacks (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M3.04 — Countermeasures against Linux attacks (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M3.04 — Countermeasures against Linux attacks (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M3.04 — Countermeasures against Linux attacks (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 — Countermeasures against Linux attacks (3 hours)?
- How would you distinguish M3.04 — Countermeasures against Linux attacks (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.04 — Countermeasures against Linux attacks (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 — Countermeasures against Linux attacks (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M3.04 — Countermeasures against Linux attacks (3 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels concept- -

Module summary: Operating-system security requires component knowledge, secure configuration, continuous patching, monitoring, and recovery readiness.

OFFICIAL MODULE 4

Web-Application and Wireless-Hacking Countermeasures

This module introduces web applications and wireless networks as systems with components, attack surfaces, tools, and defensive controls. Exercises must remain within authorised environments.

Hours: 13

CO4 · Bloom level: Understanding

Handbook study routine: Complete Module 4 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

Official source coverage for Module 4

Source basis: Official topic-row context. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

[View extracted official source transcript](#)

M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures
M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless
M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks
M4.04 3 Understanding attacks Series Test - II 1

Point-by-point study guide

– **Official syllabus point 1: M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures**

Exact source point

Syllabus text: M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures

How to understand and study it

This point is an official syllabus requirement: “M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.01 Summarize web application, components 3 Understanding Explain web application vulnerabilities, M4.02 5 Understanding counter measures ് technical terms English-് ്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures using the official terminology retained above.
- Organise the important elements of M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures into a logical sequence, classification, structure, or calculation.
- Connect M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5

Understanding counter measures is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures?
- How would you distinguish M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures from a closely related idea?
- Where would an engineer or technician encounter M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures, and what must be checked before using it?

- What common mistake would make an answer or operation involving M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.01 Summarize web application and components 3 Understanding Explain web application vulnerabilities and M4.02 5 Understanding counter measures
 താഴെ പറയുന്ന വിഷയം
 വിശദീകരിക്കുക
 exact technical term
 ഉപയോഗിക്കുക.
 definition
 principle, named parts/steps, application, limitation
 ഉപയോഗിക്കുക
 ഉപയോഗിക്കുക.
 English terminology, symbols, units, diagram labels
 ഉപയോഗിക്കുക.
 Exam answer
 concept-
 ഉപയോഗിക്കുക-
 ഉപയോഗിക്കുക
 ഉപയോഗിക്കുക.

– **Official syllabus point 2: M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless**

Exact source point

Syllabus text: M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless

How to understand and study it

This point is an official syllabus requirement: “M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.02 5 Understanding counter measures M4.03 List wireless network components, its use 2 Understanding Summarize the counter measures for wireless ് technical terms English-് ്. ് definition ് principle ്; ് ് term-് function, difference, application ് ്. Diagram, sequence, formula, unit ് ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless using the official terminology retained above.
- Organise the important elements of M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless into a logical sequence, classification, structure, or calculation.
- Connect M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless matters because an automated or digital system

must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless?
- How would you distinguish M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless from a closely related idea?
- Where would an engineer or technician encounter M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 5 Understanding counter measures M4.03 List wireless network components

and its use 2 Understanding Summarize the counter measures for wireless incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M4.02 5 Understanding counter measures M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless താഴെ പറയുന്ന വിഷയം കൃത്യമായ technical term ഉപയോഗിച്ച്. താഴെ പറയുന്ന definition ഉപയോഗിച്ച് principle, named parts/steps, application, limitation തുടങ്ങിയവ ഉൾപ്പെടെ. English terminology, symbols, units, diagram labels ഉപയോഗിച്ച്. Exam answer താഴെ പറയുന്ന concept-കൾ ഉപയോഗിച്ച്-താഴെ പറയുന്ന ഉത്തരം ഉപയോഗിച്ച്.

– **Official syllabus point 3: M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks**

Exact source point

Syllabus text: M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks

How to understand and study it

This point is an official syllabus requirement: “M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.03 List wireless network components, its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks ് technical terms English-൬. ് definition ് principle ്; ് term-൬ function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks using the official terminology retained above.
- Organise the important elements of M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks into a logical sequence, classification, structure, or calculation.
- Connect M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a

signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks?
- How would you distinguish M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks from a closely related idea?
- Where would an engineer or technician encounter M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 List wireless network components and its use 2 Understanding Summarize the counter measures for wireless M4.04 3 Understanding attacks incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M4.03 List wireless network components and its use
2 Understanding Summarize the counter measures for wireless M4.04 3
Understanding attacks താഴെ topic നൽകിയിരിക്കുന്നത് ഉപയോഗിച്ച് exact technical
term നൽകിയിരിക്കുന്നു. താഴെ definition നൽകിയിരിക്കുന്നത് principle, named
parts/steps, application, limitation നൽകിയിരിക്കുന്നു. English
terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്നു. Exam
answer നൽകിയിരിക്കുന്നത് concept-നൽകിയിരിക്കുന്നു. താഴെ-നൽകിയിരിക്കുന്നു. താഴെ നൽകിയിരിക്കുന്നു.

Exact source point

Syllabus text: M4.04 3 Understanding attacks Series Test - II 1

How to understand and study it

This point is an official syllabus requirement: “M4.04 3 Understanding attacks Series Test - II 1”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

Malayalam support: ് syllabus point ് M4.04 3 Understanding attacks Series Test - II 1 ് technical terms English-്. ് definition ് principle ്; ് term-് function, difference, application ്. Diagram, sequence, formula, unit ് revision ്.

Exam focus: Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

Handbook treatment

M4.04 3 Understanding attacks Series Test – II 1 is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.04 3 Understanding attacks Series Test – II 1 using the official terminology retained above.
- Organise the important elements of M4.04 3 Understanding attacks Series Test – II 1 into a logical sequence, classification, structure, or calculation.
- Connect M4.04 3 Understanding attacks Series Test – II 1 with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.04 3 Understanding attacks Series Test – II 1 with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 3 Understanding attacks Series Test – II 1. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.04 3 Understanding attacks Series Test – II 1 is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.04 3 Understanding attacks Series Test – II 1, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 3 Understanding attacks Series Test – II 1, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 3 Understanding attacks Series Test – II 1?
- How would you distinguish M4.04 3 Understanding attacks Series Test – II 1 from a closely related idea?
- Where would an engineer or technician encounter M4.04 3 Understanding attacks Series Test – II 1, and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 3 Understanding attacks Series Test – II 1 incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.04 3 Understanding attacks Series Test - II 1

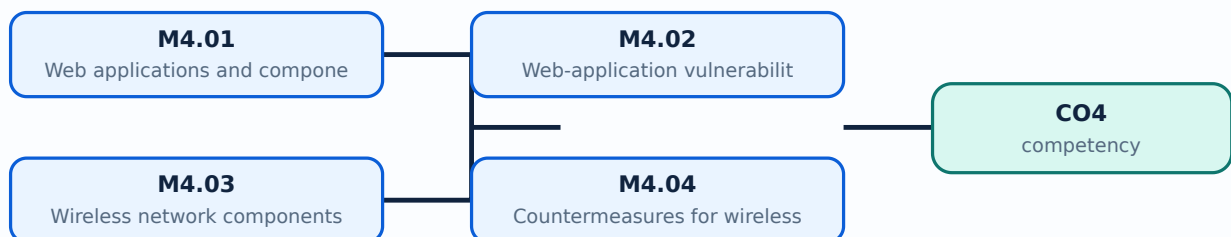
topic exact technical term . definition principle, named parts/steps, application, limitation . English terminology, symbols, units, diagram labels . Exam answer concept- -

Learning goals

- Identify web-application components and vulnerability classes.
- Explain countermeasures for web attacks and wireless attacks.
- List wireless components, wardriving concepts, and safe security practices.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

– M4.01 — Web applications and components (3 hours)

Concept and explanation

A web application commonly includes a client browser, web server, application logic, database or other backend services, session and identity components, APIs, and network controls. Requests pass through several trust boundaries, so input, authentication, authorisation, session management, output encoding, and error handling are security concerns. A web server hosts or proxies content; the application layer applies business logic; the database stores application state.

Malayalam support: Web application client, web server, application logic, database, session, identity, API ഘടകങ്ങൾ. trust boundary-ഇൻപുട്ട്, authentication, authorisation, session security ഘടകങ്ങൾ.

Study points

- Draw the request path from browser to web server, application, and database.
- Distinguish authentication from authorisation.
- Identify trust boundaries and sensitive data flows.

Engineering / workplace application: Secure development teams review web components during design, code review, testing, deployment, and monitoring.

Illustrative example: A web request may authenticate a user, then the application must separately check whether that user may read the requested record.

Common mistake: A login check does not by itself authorise every object or function requested by the user.

Handbook treatment

M4.01 — Web applications and components (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.01 — Web applications and components (3 hours) using the official terminology retained above.
- Organise the important elements of M4.01 — Web applications and components (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.01 — Web applications and components (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.01 — Web applications and components (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.01 — Web applications and components (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.01 — Web applications and components (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.01 — Web applications and components (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.01 — Web applications and components (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 — Web applications and components (3 hours)?
- How would you distinguish M4.01 — Web applications and components (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.01 — Web applications and components (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 — Web applications and components (3 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.01 — Web applications and components (3 hours) താഴെ
topic നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും exact technical term ഉപയോഗിച്ച് definition
നൽകിയിരിക്കുന്ന principle, named parts/steps, application, limitation നൽകിയിരിക്കുന്ന
തത്വം നൽകിയിരിക്കുന്നു. English terminology, symbols, units, diagram labels നൽകിയിരിക്കുന്നു.
Exam answer നൽകിയിരിക്കുന്ന concept-നൽകിയിരിക്കുന്നവയിൽ ഏതെങ്കിലും
തത്വം നൽകിയിരിക്കുന്നു.

Concept and explanation

Web vulnerabilities arise from unvalidated input, injection, broken authentication or session management, missing authorisation, insecure direct object access, cross-site scripting, insecure file handling, weak configuration, exposed secrets, and unsafe error messages. Countermeasures include allow-list validation, parameterised queries, output encoding, secure session cookies, MFA, least privilege, access-control checks on every request, secure headers, patching, secrets management, logging, testing, and safe error handling.

Malayalam support: Web vulnerabilities input validation, injection, authentication, session, authorisation, XSS, file handling, configuration, secrets സെക്രറ്റുകൾ, പാർമിറ്ററൈസ്ഡ് ക്വെറികൾ, ഔട്ട്പുട്ട് എൻകോഡിംഗ്, അക്സസ് ചെക്കുകൾ, സെക്യൂർ കൂക്കികൾ, പാച്ചിംഗ്, സെക്രറ്റ്സ് മാനേജ്മെന്റ്, കൗണ്ടർമേചേർസ്.

Study points

- For each vulnerability class, state the failing control and a defensive measure.
- Keep authentication, authorisation, and input validation conceptually separate.
- Use logging and testing to verify controls.

Engineering / workplace application: Secure web applications protect customer data, APIs, administrative portals, and online services.

Illustrative example: A server validates an account identifier against the authenticated user and uses a parameterised query instead of trusting a browser-supplied value.

Common mistake: Client-side validation alone is not a security control because an attacker can send requests directly.

Handbook treatment

M4.02 — Web-application vulnerabilities and countermeasures (5 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

Learning objectives

- Define and scope M4.02 — Web-application vulnerabilities and countermeasures (5 hours) using the official terminology retained above.
- Organise the important elements of M4.02 — Web-application vulnerabilities and countermeasures (5 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.02 — Web-application vulnerabilities and countermeasures (5 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.02 — Web-application vulnerabilities and countermeasures (5 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.02 — Web-application vulnerabilities and countermeasures (5 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

Engineering application lens

The engineering value of M4.02 — Web-application vulnerabilities and countermeasures (5 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

Worked answer method

Given: the topic, operating condition, diagram, data, or situation stated in the question.

Required: definition, explanation, comparison, procedure, calculation, or application as requested.

Method: organise the answer under principle, named elements, sequence or relationship, and application.

Answer: use the requested technical terms, units, labels, assumptions, and a concluding statement.

Exam answer blueprint

For a short-answer question on M4.02 — Web-application vulnerabilities and countermeasures (5 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.02 — Web-application vulnerabilities and countermeasures (5 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 — Web-application vulnerabilities and countermeasures (5 hours)?
- How would you distinguish M4.02 — Web-application vulnerabilities and countermeasures (5 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.02 — Web-application vulnerabilities and countermeasures (5 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 — Web-application vulnerabilities and countermeasures (5 hours) incorrect?

Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

Malayalam support: M4.02 — Web-application vulnerabilities and countermeasures (5 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ച് കൃത്യമായ technical term ഉപയോഗിക്കുക. definition ഉൾപ്പെടെ principle, named parts/steps, application, limitation തുടങ്ങിയവ ഉൾപ്പെടെ. English terminology, symbols, units, diagram labels ഉൾപ്പെടെ ഉപയോഗിക്കുക. Exam answer concept-ബോക്സ് ഉപയോഗിച്ച്-എല്ലാ ചോദ്യങ്ങൾക്കും ഉത്തരം നൽകുക.

— M4.03 — Wireless network components and their use (2 hours)

Concept and explanation

Wireless networks include clients, access points, wireless controllers, antennas, radio channels, authentication services, encryption, and wired backhaul. The security posture depends on configuration, coverage, channel use, identity, encryption, guest isolation, firmware, and monitoring. Wireless clients and access points exchange frames over a medium that can be observed beyond physical walls, so strong authentication and encryption are important.

Malayalam support: Wireless network- client, access point, controller, antenna, channel, authentication, encryption, wired backhaul. Radio signal physical boundary-; strong authentication and encryption.

Study points

- Identify the role of client, access point, controller, authentication, and encryption.
- Explain why radio coverage affects security.
- Separate guest and administrative access where appropriate.

Engineering / workplace application: Enterprise wireless design uses identity, segmentation, monitoring, and controlled configuration.

Illustrative example: A guest SSID can be isolated from internal systems while internet access is monitored and rate-limited.

Common mistake: A hidden network name does not replace encryption and authentication.

Handbook treatment

M4.03 — Wireless network components and their use (2 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

Learning objectives

- Define and scope M4.03 — Wireless network components and their use (2 hours) using the official terminology retained above.
- Organise the important elements of M4.03 — Wireless network components and their use (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.03 — Wireless network components and their use (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.03 — Wireless network components and their use (2 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.03 — Wireless network components and their use (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

Engineering application lens

For engineering practice, M4.03 — Wireless network components and their use (2 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

Worked answer method

Given: the input conditions, required output, and any stated constraints.

Required: the logic sequence, program structure, truth table, or operating result.

Method: break the problem into named states or steps; evaluate them in order; test at least one alternate condition.

Answer: show the final sequence and state the expected output for each important condition.

Exam answer blueprint

For a short-answer question on M4.03 — Wireless network components and their use (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.03 — Wireless network components and their use (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 — Wireless network components and their use (2 hours)?
- How would you distinguish M4.03 — Wireless network components and their use (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.03 — Wireless network components and their use (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 — Wireless network components and their use (2 hours) incorrect?

Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

Malayalam support: M4.03 — Wireless network components and their use (2 hours) താഴെ പറയുന്ന വിഷയം സംബന്ധിച്ചുള്ള കൃത്യമായ technical term ഉപയോഗിക്കുക. definition, principle, named parts/steps, application, limitation എന്നിവ ഉൾക്കൊള്ളിക്കുക. English terminology, symbols, units, diagram labels ഉപയോഗിക്കുക. Exam answer concept-പ്രകാരം വിശദീകരിക്കുക.

– M4.04 — Countermeasures for wireless attacks (3 hours)

Concept and explanation

Wireless countermeasures include modern encryption and authentication, strong credentials, protected management frames where supported, updated firmware, segmentation, guest isolation, rogue-access-point detection, secure configuration, monitoring, and user awareness. Wardriving is the process of surveying wireless signals while moving through an area; it may be used in authorised security assessment but can expose sensitive network information. Tools for wireless assessment must be used only in a permitted lab or engagement.

Malayalam support: Wireless security- strong authentication/encryption, firmware updates, segmentation, guest isolation, rogue AP detection, monitoring ഏകദേശം പരിശോധന. Wardriving authorised survey പരമ പരിശോധന പരിശോധന.

Study points

- Explain why encryption, authentication, segmentation, and monitoring work together.
- Define wardriving at a conceptual level.
- State the authorisation requirement for wireless tools.

Engineering / workplace application: Wireless assessments help organisations find weak coverage, legacy encryption, rogue access points, and unsafe guest-network paths.

Illustrative example: A security team can map authorised coverage, identify a rogue access point, disable it, rotate credentials, and review logs.

Common mistake: Changing the SSID name is not an effective substitute for secure encryption and access control.

Handbook treatment

M4.04 — Countermeasures for wireless attacks (3 hours) should be followed as a signal or information path. Explain what is generated, how it is modified or transmitted, what can disturb it, and how the receiver reconstructs or uses it.

Learning objectives

- Define and scope M4.04 — Countermeasures for wireless attacks (3 hours) using the official terminology retained above.
- Organise the important elements of M4.04 — Countermeasures for wireless attacks (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.04 — Countermeasures for wireless attacks (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Web-Application and Wireless-Hacking Countermeasures.
- Write an examination answer on M4.04 — Countermeasures for wireless attacks (3 hours) with a clear opening, technical middle, and final application or limitation.

Conceptual framework

Use the following frame while reading M4.04 — Countermeasures for wireless attacks (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the information-bearing signal and the relevant source or channel.
- Describe the blocks or stages in order.
- Explain the role of bandwidth, noise, attenuation, distortion, coding, modulation, or protocol rules where applicable.
- Compare performance using range, capacity, reliability, complexity, cost, and application.

Engineering application lens

Engineering systems use M4.04 — Countermeasures for wireless attacks (3 hours) when information must be transferred reliably between equipment, instruments, controllers, or users. The choice of method is a balance between signal quality, distance, data rate, interference, infrastructure, safety, and maintenance.

Worked answer method

Given: source, channel, receiver, signal condition, or communication requirement.

Required: block diagram, operating explanation, comparison, or fault analysis.

Method: trace the signal from source to destination and mark where conversion, loss, interference, or recovery occurs.

Answer: state the principle, blocks, performance factors, application, and limitation.

Exam answer blueprint

For a short-answer question on M4.04 — Countermeasures for wireless attacks (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

Self-check questions

- What is M4.04 — Countermeasures for wireless attacks (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 — Countermeasures for wireless attacks (3 hours)?
- How would you distinguish M4.04 — Countermeasures for wireless attacks (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.04 — Countermeasures for wireless attacks (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 — Countermeasures for wireless attacks (3 hours) incorrect?

Common mistakes and safety emphasis

- Using transmitter, channel, and receiver as labels without explaining their functions.
- Confusing bandwidth, data rate, frequency, and range.
- Ignoring noise, attenuation, interference, synchronization, or protocol compatibility.
- Comparing systems without using common criteria.

Malayalam support: M4.04 — Countermeasures for wireless attacks (3 hours)

topic exact technical term definition principle, named parts/steps, application, limitation English terminology, symbols, units, diagram labels concept- -

Module summary: Web and wireless security both require component awareness, strong identity, least privilege, secure protocols, monitoring, and authorised testing.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

ning-map diagram.

ning-map diagram.

ing-map diagram.

Open the module to view its labelled learning-map diagram.

Module 2: Footprinting,

Module 3: Operating-

Module 4: Web-Application

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Source anomaly: The supplied Course Outcomes table lists Series Test as 2 hours, while the course outline separately lists Series Test I and Series Test II as 1 hour within Modules 2 and 4; both official entries are preserved.
- No separate CIE/ESE mark distribution is stated in the supplied PDF.

Module-wise Questions and Answers

module-1 — Hacking, Malware, Network Attacks and Countermeasures

1. Explain Hacking and ethical hacking. [3 marks]

Hacking is the use of technical methods to access, manipulate, or test systems; the intent and authorisation determine whether an activity is malicious, unauthorised, or ethical. Ethical hacking is a controlled security assessment performed within written scope, rules of engagement, and reporting requirements. Penetration testing simulates authorised attack paths to identify exploitable weaknesses and verify controls.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Ethical issues related to hacking. [3 marks]

Ethical issues include consent, privacy, data minimisation, scope control, safe handling of credentials, non-disruption, responsible disclosure, evidence protection, and avoiding unauthorised persistence. A tester must stop when the scope is exceeded, protect discovered personal data, and report critical findings promptly. Laws and organisational policies apply even when a vulnerability is easy to exploit.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Malware classes, actions and protection. [3 marks]

Malware is malicious software designed to disrupt, damage, spy, extort, or gain unauthorised access. Viruses attach to files and spread when executed; worms self-propagate across networks; Trojans masquerade as legitimate software; spyware monitors activity; adware displays unwanted advertising and may track behaviour. Protection includes patching, least privilege, endpoint protection, secure backups, application control, user awareness, network segmentation, and monitoring.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Network attacks. [3 marks]

The syllabus includes DoS, DDoS, buffer overflow, ping of death, session hijacking, brute-force, man-in-the-middle, dictionary, and replay attacks. DoS exhausts a service resource; DDoS distributes the traffic or workload across many sources. A buffer overflow abuses inadequate bounds checking. Session hijacking steals or reuses session state. Brute-force and dictionary attacks try credentials; MITM intercepts or alters communication; replay reuses captured valid messages.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Footprinting, Social Engineering and Port Scanning

5. Explain Footprinting and tools used for it. [3 marks]

Footprinting is the systematic collection of information about an organisation, domain, network, technology stack, and public exposure before a security assessment. Passive

methods use public records and observable information; active methods query or interact with systems and require explicit authorisation. Web search, domain records, certificate information, DNS data, and approved assessment tools can reveal attack surface. Findings should be validated and documented with source and time.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Competitive intelligence methods. [3 marks]

Competitive intelligence in security assessment means structured analysis of publicly available information about technologies, suppliers, domains, job postings, exposed services, and organisational change. The information should be collected lawfully, evaluated for reliability, and used to understand likely technology and risk rather than to target individuals. Corroboration is important because public data may be stale or misleading.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Social-engineering methods. [3 marks]

Social engineering manipulates people rather than software. Shoulder surfing observes information directly; dumpster diving searches discarded material; piggybacking follows an authorised person into a restricted area; phishing uses deceptive messages or pages to obtain action or information. Awareness, verification, physical access control, secure disposal, MFA, reporting channels, and least privilege reduce risk. Ethical simulations need consent, safety boundaries, and a clear learning objective.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Port scanning, methods and tools. [3 marks]

Port scanning probes network endpoints to identify reachable ports and infer services. TCP connect scans complete a connection; SYN scans examine handshake responses without completing a full connection in the usual case; UDP scans require different interpretation because responses may be absent or filtered. Ping sweeps identify responding hosts. The syllabus names Nmap and Nessus as tools and also includes crafting IP packets and spoofing. Tool output is an observation that must be validated, documented, and handled within scope.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Operating-System Vulnerabilities and Hardening

9. Explain Windows vulnerable components. [3 marks]

The syllabus lists Windows file systems, Windows RPC, NetBIOS, Server Message Block, Common Internet File System, null sessions, web services, passwords, and authentication. Vulnerabilities may arise from legacy protocols, excessive exposure, weak permissions, unpatched services, insecure shares, or weak credentials. Assessment should identify the component, exposure, evidence, impact, and safe remediation rather than merely naming a tool.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Methods for hardening Windows. [3 marks]

Windows hardening includes timely patching, removal or restriction of unnecessary services, secure configuration of file sharing and RPC, strong password and MFA policy, least privilege, host firewall, endpoint protection, logging, secure remote administration, backup, and vulnerability verification. Hardening is a baseline-and-monitor process, not a one-time switch. Changes should be documented and tested.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Linux OS vulnerabilities. [3 marks]

Linux vulnerabilities can involve unpatched kernels or packages, weak file permissions, insecure services, exposed management interfaces, unsafe scripts, weak authentication, misconfigured sudo, and vulnerable web or network daemons. Identification uses configuration review, package and service inventories, logs, approved scanners, and manual verification. The operating system name alone does not determine security; configuration and maintenance matter.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Countermeasures against Linux attacks. [3 marks]

Linux countermeasures include patch management, minimal services, secure SSH configuration, key-based authentication, least privilege, firewalling, file-integrity monitoring, mandatory access controls where appropriate, backups, logging, alerting, and tested incident response. Administrators should separate service accounts, protect secrets, and review sudo rules. Countermeasures must be matched to the threat and operational requirement.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Web-Application and Wireless-Hacking Countermeasures

13. Explain Web applications and components. [3 marks]

A web application commonly includes a client browser, web server, application logic, database or other backend services, session and identity components, APIs, and network controls. Requests pass through several trust boundaries, so input, authentication, authorisation, session management, output encoding, and error handling are security concerns. A web server hosts or proxies content; the application layer applies business logic; the database stores application state.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Web-application vulnerabilities and countermeasures. [3 marks]

Web vulnerabilities arise from unvalidated input, injection, broken authentication or session management, missing authorisation, insecure direct object access, cross-site scripting, insecure file handling, weak configuration, exposed secrets, and unsafe error messages. Countermeasures include allow-list validation, parameterised queries, output encoding, secure session cookies, MFA, least privilege, access-control checks on every request, secure headers, patching, secrets management, logging, testing, and safe error handling.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Wireless network components and their use. [3 marks]

Wireless networks include clients, access points, wireless controllers, antennas, radio channels, authentication services, encryption, and wired backhaul. The security posture depends on configuration, coverage, channel use, identity, encryption, guest isolation, firmware, and monitoring. Wireless clients and access points exchange frames over a medium that can be observed beyond physical walls, so strong authentication and encryption are important.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Countermeasures for wireless attacks. [3 marks]

Wireless countermeasures include modern encryption and authentication, strong credentials, protected management frames where supported, updated firmware, segmentation, guest isolation, rogue-access-point detection, secure configuration, monitoring, and user awareness. Wardriving is the process of surveying wireless signals while moving through an area; it may be used in authorised security assessment but can expose sensitive network information. Tools for wireless assessment must be used only in a permitted lab or engagement.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. Define or explain *Hacking and ethical hacking* with one relevant application. (5 marks)
2. Define or explain *Ethical issues related to hacking* with one relevant application. (5 marks)
3. Define or explain *Footprinting and tools used for it* with one relevant application. (5 marks)
4. Define or explain *Competitive intelligence methods* with one relevant application. (5 marks)
5. Define or explain *Windows vulnerable components* with one relevant application. (5 marks)

6. **6.** Define or explain *Methods for hardening Windows* with one relevant application. (5 marks)
7. **7.** Define or explain *Web applications and components* with one relevant application. (5 marks)
8. **8.** Define or explain *Web-application vulnerabilities and countermeasures* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Hacking, Malware, Network Attacks and Countermeasures:** Ethical practice begins with authorisation and continues through layered controls, safe testing, and responsible reporting.
- **Footprinting, Social Engineering and Port Scanning:** Reconnaissance is evidence gathering, not exploitation: scope, safety, validation, and documentation control every demonstration.
- **Operating-System Vulnerabilities and Hardening:** Operating-system security requires component knowledge, secure configuration, continuous patching, monitoring, and recovery readiness.
- **Web-Application and Wireless-Hacking Countermeasures:** Web and wireless security both require component awareness, strong identity, least privilege, secure protocols, monitoring, and authorised testing.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Hacking and ethical hacking	Hacking is the use of technical methods to access, manipulate, or test systems; the intent and authorisation determine whether an activity is malicious, unauthorised, or ethical. Ethical hacking is a controlled security assessment performed within written scop
Ethical issues related to hacking	Ethical issues include consent, privacy, data minimisation, scope control, safe handling of credentials, non-disruption, responsible disclosure, evidence protection, and avoiding unauthorised persistence. A tester must stop when the scope is exceeded, protect
Malware classes, actions and protection	Malware is malicious software designed to disrupt, damage, spy, extort, or gain unauthorised access. Viruses attach to files and spread when executed; worms self-propagate across networks; Trojans masquerade as legitimate software; spyware monitors activity; a
Network attacks	The syllabus includes DoS, DDoS, buffer overflow, ping of death, session hijacking, brute-force, man-in-the-middle, dictionary, and replay attacks. DoS exhausts a service resource; DDoS distributes the traffic or workload across many sources. A buffer overflow
Countermeasures for network attacks	Countermeasures are layered: secure configuration and patching, firewalls and segmentation, strong authentication and MFA, encrypted protocols, input validation, rate limiting, anti-DDoS services, secure session management, logging, alerting, backups, and inci
Footprinting and tools used for it	Footprinting is the systematic collection of information about an organisation, domain, network, technology stack, and public exposure before a security assessment. Passive methods use public records and observable information; active methods query or interact
Competitive intelligence methods	Competitive intelligence in security assessment means structured analysis of publicly available information about technologies, suppliers, domains, job postings, exposed services, and organisational change. The information should be collected lawfully, evaluat
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Official References and Resources

References listed in the supplied syllabus

1. Michael T. Simpson, Kent Backman and James Corley, Hands on Ethical Hacking and Network Defense, 2nd Edition.
2. W. Stallings, Data and Computer Communications.

3. W. Stallings, Cryptography and Network Security: Principles and Practice.

Official learning resources listed

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In-page Search

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